Example 8.12

Find $(\vec{a}+3\vec{b})\cdot(2\vec{a}-\vec{b})$ if $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b}=3\hat{i}+2\hat{j}-\hat{k}$

Solution

$$(\vec{a}+3\vec{b})\cdot(2\vec{a}-\vec{b})=2\vec{a}\cdot\vec{a}+5\vec{a}\cdot\vec{b}-3\vec{b}\cdot\vec{b}=2(1+1+4)+5(3+2-2)-3(9+4+1)=-15\,.$$

Example 8.13

If $\vec{a}=2\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=-\hat{i}+2\hat{j}+\hat{k}$ and $\vec{c}=3\hat{i}+\hat{j}$ be such that $\vec{a}+\lambda\vec{b}$ is perpendicular to $\vec{c}$ then find λ.

Solution

$$\begin{aligned}(\vec{a}+\lambda\vec{b})\cdot\vec{c}=0 &\Rightarrow \vec{a}\cdot\vec{c}+\lambda\vec{b}\cdot\vec{c}=0\\ &\Rightarrow (6+2)+\lambda(-3+2)=0\\ &\Rightarrow \lambda=8.\end{aligned}$$

Example 8.14

If $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|$ prove that $\vec{a}$ and $\vec{b}$ are perpendicular.

Solution

$$\begin{aligned}|\vec{a}+\vec{b}| &= |\vec{a}-\vec{b}|\\ |\vec{a}+\vec{b}|^2 &= |\vec{a}-\vec{b}|^2\\ |\vec{a}|^2+|\vec{b}|^2+2\vec{a}\cdot\vec{b} &= |\vec{a}|^2+|\vec{b}|^2-2\vec{a}\cdot\vec{b}\\ \Rightarrow\quad 4\vec{a}\cdot\vec{b} &= 0\\ \vec{a}\cdot\vec{b} &= 0\end{aligned}$$

Hence $\vec{a}$ and $\vec{b}$ are perpendicular.

Example 8.15

For any vector $\vec{r}$ prove that $\vec{r}=(\vec{r}\cdot\hat{i})\,\hat{i}+(\vec{r}\cdot\hat{j})\hat{j}+(\vec{r}\cdot\hat{k})\hat{k}$.

Solution

Let $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$

$$\begin{aligned}\vec{r}\cdot\hat{i} &= \left(x\hat{i}+y\hat{j}+z\hat{k}\right)\cdot\hat{i}=x\\ \vec{r}\cdot\hat{j} &= \left(x\hat{i}+y\hat{j}+z\hat{k}\right)\cdot\hat{j}=y\\ \vec{r}\cdot\hat{k} &= \left(x\hat{i}+y\hat{j}+z\hat{k}\right)\cdot\hat{k}=z\end{aligned}$$

$$(\vec{r}\cdot\hat{i})\,\hat{i}+(\vec{r}\cdot\hat{j})\hat{j}+(\vec{r}\cdot\hat{k})\hat{k}=x\hat{i}+y\hat{j}+z\hat{k}=\vec{r}$$

Thus $\vec{r}=(\vec{r}\cdot\hat{i})\hat{i}+(\vec{r}\cdot\hat{j})\hat{j}+(\vec{r}\cdot\hat{k})\hat{k}$.

Example 8.16

Find the angle between the vectors $5\hat{i}+3\hat{j}+4\hat{k}$ and $6\hat{i}-8\hat{j}-\hat{k}$.

Solution

Let $\vec{a}=5\hat{i}+3\hat{j}+4\hat{k}$, and $\vec{b}=6\hat{i}-8\hat{j}-\hat{k}$.

Let θ be the angle between them.

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